



Enterprise Virtualization with VMware ESXi

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Introduction

Enterprise virtualization is an essential part of modern IT infrastructure, allowing organizations to efficiently use physical computing resources by creating and managing virtual environments. VMware ESXi provides a practical platform for hosting and administering virtual machines while managing resources such as CPU, memory, storage, and networking.

Objective

To understand the purpose of virtualization in modern IT environments and perform basic administrative operations on an ESXi host.

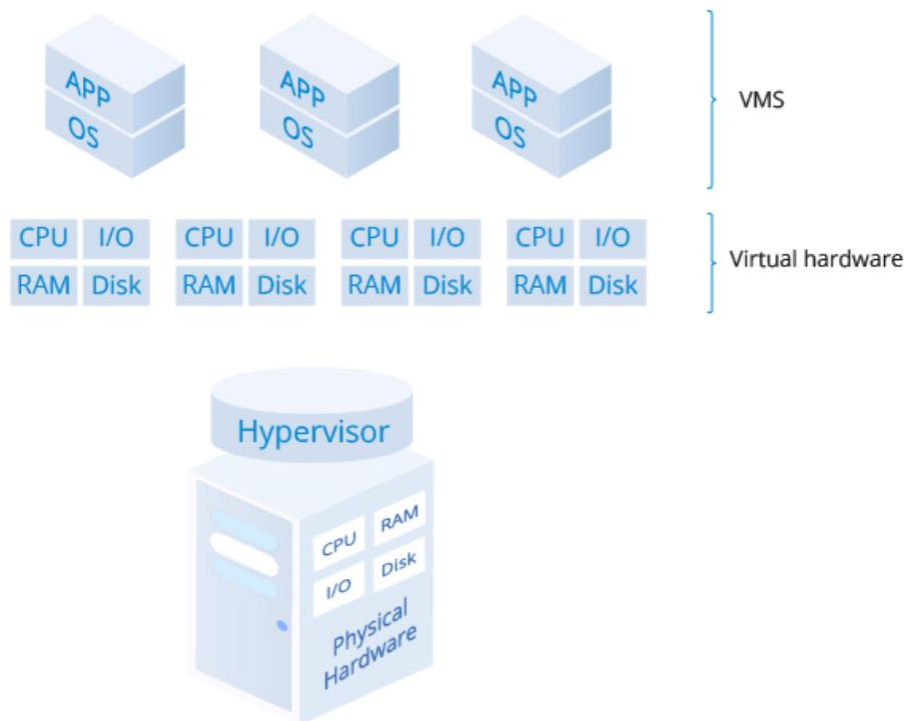
Theory Topics

Fundamentals of Enterprise Virtualization

- Physical vs virtual infrastructure

Physical infrastructure refers to servers, storage, and networking equipment that you own, deploy, and maintain directly. This is the foundation behind traditional data centers and on-premises environments.

Virtual infrastructure, in contrast, uses software to simulate and manage these resources. A single physical machine can run multiple virtual machines, each isolated but sharing the same underlying hardware.

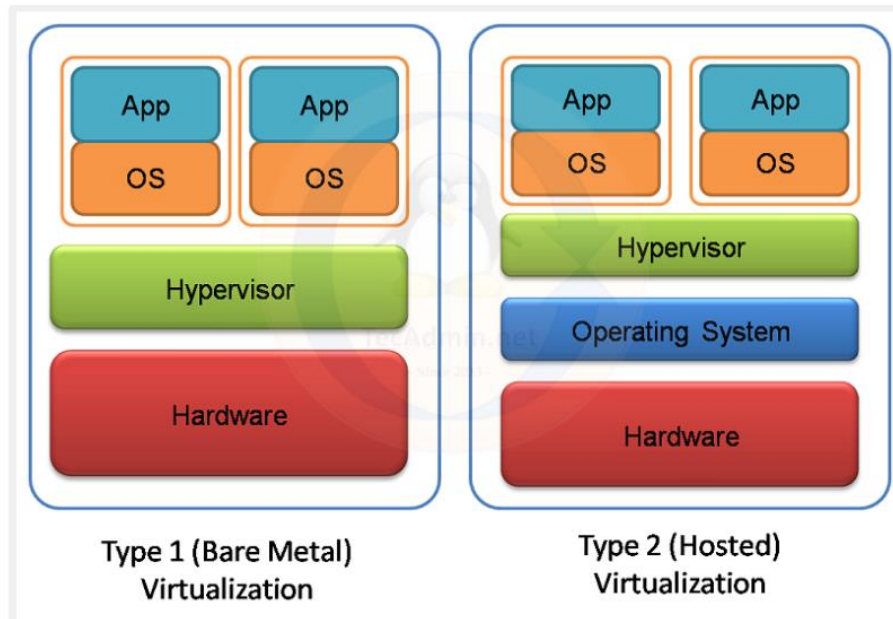


- Importance & Benefits of enterprise virtualization
 - Better flexibility and scalability
 - Improved disaster recovery and continuous business
 - Increased IT efficiency and productivity
 - Better collaboration and remote work support
 - Reduced downtime and maintenance costs
- Challenges of enterprise virtualization
 - Migrating from legacy systems
 - Performance overhead
 - Data privacy and compliance concerns
 - Need for skilled IT professionals
 - Managing hybrid or multi-cloud environments
- Enterprise use cases of virtualization
 - Network and Storage Infrastructure Abstraction
 - Legacy Application Hosting
 - Secure Remote Workforces

- Automated Software Testing & CI/CD Pipelines
- Live Workload Migration & Hardware Maintenance
- Cybersecurity Sandboxing

Hypervisor Technologies

- Type-1 vs Type-2 hypervisors



Feature	Type 1 Hypervisors	Type 2 Hypervisors
Runs On	Physical hardware directly	Host operating system
Performance	Efficient	Lower due to host OS overhead
Security	Stronger (smaller attack surface)	Weaker (relies on host OS)
Setup Complexity	Higher	Simple, user-friendly
HCI Integration	Commonly used	Rarely used
Examples	VMware ESXi, Microsoft Hyper-V, KVM	Oracle VirtualBox, VMware Workstation Pro, Parallels Desktop

- VMware ESXi
 - Enterprise-class, type-1 hypervisors (bare-metal)

- Small and efficient: Bare-metal design with minimal resource requirements
- Scalable performance: Supports up to 960 logical CPUs and 16 TB of memory per ESXi host in vSphere 8.0 Update 2 (or up to 896 logical CPUs with 24 TB of memory)
- Security-first architecture: Logging, auditing, VM encryption, and role-based access controls protect workloads
- Intuitive management: Admins can use REST APIs, the vSphere CLI, or automation tools to streamline routine tasks
- Other virtualization platforms

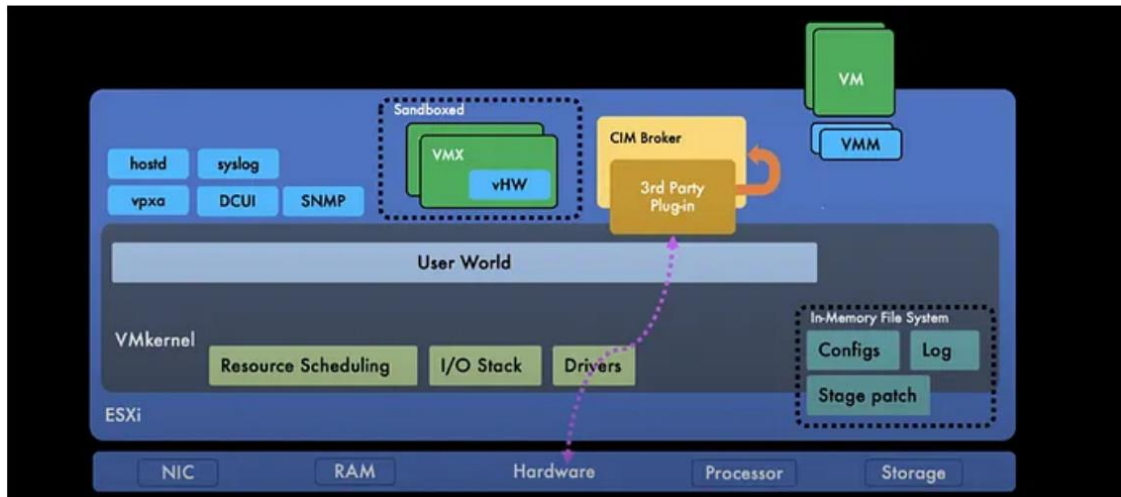
VirtualBox	hosted (Type 2) hypervisor,
OpenShift (Red Hat)	enterprise-grade Kubernetes platform with integrated virtualization
Hyper-V (Microsoft)	optimized for Windows Server-intensive workloads and designed to deliver a cost advantage for Microsoft license holders
XenServer (Citrix)	high-performance virtualization platform for graphics-intensive workloads

VMware Virtualization Ecosystem

ESXi	Type-1 bare-metal hypervisor that partitions physical server hardware into isolated virtual machines (VMs).
vCenter	(server) Provides centralized management, monitoring, and automation across multiple ESXi hosts and clusters.
vSphere	The overarching suite containing ESXi and vCenter, featuring high availability (HA), Distributed Resource Scheduler (DRS), and vMotion for live workload migration
workstation	Local hypervisors enabling developers to run multi-OS sandboxes on Windows, Linux, and Macs

VMware ESXi Architecture

- ESXi Host Architecture & VMkernel



VM Kernel: This is the core component responsible for resource scheduling, managing the I/O stack, and handling drivers. It's essential for the smooth operation of the entire system.

VMM (Virtual Machine Monitor): This creates a secure execution environment for virtual machines (VMs), allowing them to operate with virtual hardware.

VMX (Helper Process): VMX acts as a sandbox, ensuring VMs remain isolated from each other, enhancing security and stability.

User World: This component provides management access, monitors hardware, and serves as an interface to the VM Kernel, streamlining administrative tasks.

Hostd: Hostd offers programmatic (API) access to the VM kernel, managing local users and groups on ESXi. It also functions as a reverse proxy for ESXi communication.

VPXA: Responsible for facilitating communication between the vCenter and the host, VPXA uses the VPX user to relay requests from vCenter to Hostd.

Syslog: This component is crucial for logging data to a remote syslog server, aiding in monitoring and troubleshooting.

DCUI (Direct Control User Interface): DCUI provides a direct interface for system control and management.

SNMP: SNMP is used for collecting information from the host, crucial for network management and monitoring.

Virtual Machine Management

- **Virtual Disk:** A virtual disk is the VM's "hard drive." Instead of a physical HDD/SSD being directly attached to the VM, ESXi provides a virtual disk file, usually a VMDK (Virtual Machine Disk).
- **Snapshots:** a point-in-time record of a VM.
- **VM Power Operations:** actions used to control the power state of a virtual machine in VMware ESX (power on/ power off/ shut down/ restart/ reset/ suspend)

Virtual Networking in VMware ESXi

- **Virtual Networking:** Virtual networking is a technology that enables the remote control and management of computers or servers over the Internet.
- **Virtual networking components**

vSwitch : a virtual switch that is a software analog of a physical switch. Though not a hardware device, a virtual switch passes

traffic using the same concepts used in the OSI networking model as physical switches.

The difference between vSwitch and physical switch networking in the context of VMware is the absence of a table of MAC addresses for devices that the vSwitch does not directly know about, unlike physical switches.

In vSphere, there are two types of virtual switches that can be provisioned, the vSphere Standard Switch (vSS) and Distributed Switch (vDS). The vDS requires Enterprise Plus licensing to be available and is configured at the vCenter level.

Port group: A Virtual Machine Port Group in VMware vSphere is a logical grouping of virtual ports on a vSwitch that provides network connectivity for virtual machines.

It connects Virtual Machines (VMs) to the vSwitch, enabling them to communicate with other VMs and external networks.

vNIC : The virtual network adapter attached to a VM.

Physical NIC (vmnic): The actual network adapter on the physical machine running ESXi.

ESXi Storage Management

- **Virtual Storage:** Virtual storage means that each running program can assume it has access to all of the storage defined by the architecture's addressing scheme.
- **Datastore Architecture:** A Datastore is a logical storage unit that can use a disk space on physical device or span several physical devices. Datastore is used to store VMs, templates, and ISO

images. Types (VMFS, NFS, Virtual SAN 'vSAN', Virtual Volumes 'VVol')

- VMFS: a high-performance, shared file system designed for virtualization.

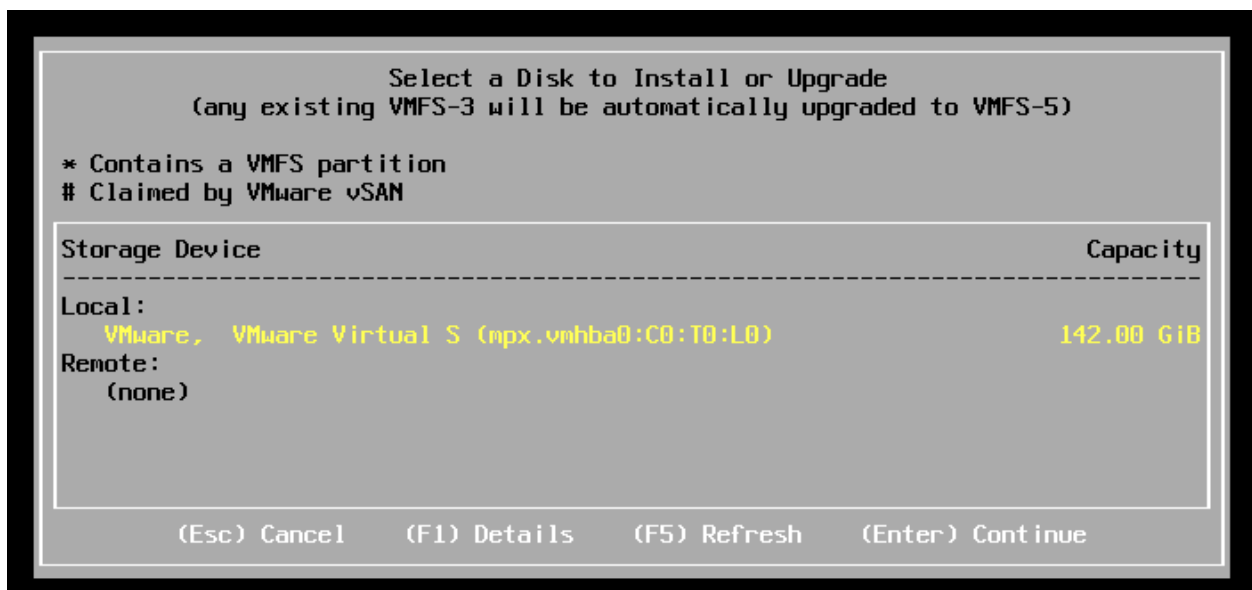
It stores: Virtual machine files (.vmx) - Virtual disks (.vmdk) - VM snapshots - VM templates - ISO files - Suspend & swap files

- ISO Image Management: an Image of a CD or DVD which you can burn, resulting in a duplicate of the original
- Virtual Disk Files (VMDK): The actual virtual disk, or contents of the virtual machine.

Implementation

Lab Environment: Installation of VMware ESXi

Installing ESXi ISO image and using it inside VMwarestation as nested virtualization.



VMware ESXi 8.0.3 (VMKernel Release Build 24677879)

VMware, Inc. VMware20,1

AMD Ryzen 5 PRO 6650U with Radeon Graphics
4 GiB Memory

To manage this host, go to:
<https://192.168.203.131/> (DHCP)
<https://1fe80::20c:29ff:fe52:3a061/> (STATIC)

Configuration of the ESXi Management Network

on VM -> F2 -> configure management network

-> IPv4 configuration (use space-bar to select option)

IPv4 Configuration

This host can obtain network settings automatically if your network includes a DHCP server. If it does not, the following settings must be specified:

- () Disable IPv4 configuration for management network
- () Use dynamic IPv4 address and network configuration
- (o) Set static IPv4 address and network configuration:

IPv4 Address	[192.168.203.135]
Subnet Mask	[255.255.255.0]
Default Gateway	[192.168.203.2]

<Up/Down> Select <Space> Mark Selected

<Enter> OK <Esc> Cancel

-> DNS configuration

DNS Configuration

This host can only obtain DNS settings automatically if it also obtains its IP configuration automatically.

() Obtain DNS server addresses and a hostname automatically
(o) Use the following DNS server addresses and hostname:

Primary DNS Server	[192.168.203.3	]
Alternate DNS Server	[	]
Hostname	[esxi-host	]

<Up/Down> Select <Space> Mark Selected <Enter> OK <Esc> Cancel

ESXi Host Client Access

Summary, Monitor, Networking, Storage, and Virtual Machines sections

The screenshot displays the ESXi Host Client interface for a host named **localhost.localdomain**. The interface is divided into several sections:

- Host Summary:** Shows the host's version (8.0 Update 3), state (Normal), and uptime (0.03 days). It also includes a status icon and a checkmark.
- Hardware:** Lists the host's hardware details, including:
 - Manufacturer: VMware, Inc.
 - Model: VMware20,1
 - CPU: 2 CPUs x AMD Ryzen 5 PRO 6650U with Radeon Graphics
 - Memory: 4 GB
 - Virtual flash: 0 B used, 0 B capacity
 - Networking:
 - Hostname: localhost.localdomain
 - IP addresses: 1. vmk0: 192.168.203.131, 2. vmk0: fe80::20c:29ff:fe52:3a06
 - DNS servers: 1. 192.168.203.2
 - Default gateway: 192.168.203.2
 - IPv6 enabled: Yes
- Configuration:** Shows the host's configuration details, including:
 - Image profile: ESXi-8.0U3e-24677879-standard (VMware, Inc.)
 - vSphere HA state: Not configured
 - vMotion: Not supported
- System Information:** Provides system details such as:
 - Date/time on host: Wednesday, August 12, 2026, 05:29:28 UTC
 - Install date: Wednesday, August 12, 2026, 04:31:00 UTC
 - Asset tag: No Asset Tag
 - Serial number: VMware-56 4d 11 94 c9 f5 29 60-b1 a8 a9 02 f9 52 3a 06
 - BIOS version: VMW201.00V.24866131.B64.2507211911
 - BIOS release date: Sunday, July 20, 2025, 20:00:00 -0400
- Performance summary last hour:** Shows a green indicator for "Consumed host CPU".
- Recent tasks:** A table at the bottom showing the history of tasks performed on the host.

Creating a Virtual Machine

New virtual machine - windows-vm (ESXi 8.0 U2 virtual machine)

- Select creation type
- Select a name and guest OS
- Select storage
- Customize settings
- Ready to complete

Name	windows-vm
Datastore	datastore1
Guest OS name	Microsoft Windows 10 (64-bit)
Compatibility	ESXi 8.0 U2 virtual machine
vCPUs	1
Memory	2000 MB
Network adapters	1
Network adapter 1 network	VM Network
Network adapter 1 type	E1000e
IDE controller 0	IDE 0
IDE controller 1	IDE 1
SCSI controller 0	LSI Logic SAS
SATA controller 0	New SATA controller
Hard disk 1	
Capacity	10 GB
Datastore	[datastore1] windows-vm/
Mode	Dependent
Provisioning	Thin provisioned

CANCEL BACK NEXT FINISH

Managing the Virtual Machine Hardware, and configuring for Better Performance

Create / Register VM | Console | Power on | Power off | Suspend | Refresh | Actions

Virtual machine	Status	Used space	Guest OS	Host name	Host CPU	Host memory
✓ windows-vm	Normal	0 B	Microsoft Windows 10 (64-bit)	Unknown	0 MHz	0 MB

Quick filters... 1 items



windows-vm

Guest OS: Microsoft Windows 10 (64-bit) VBS enabled

Compatibility: No

VMware Tools: 1

CPUs: 1

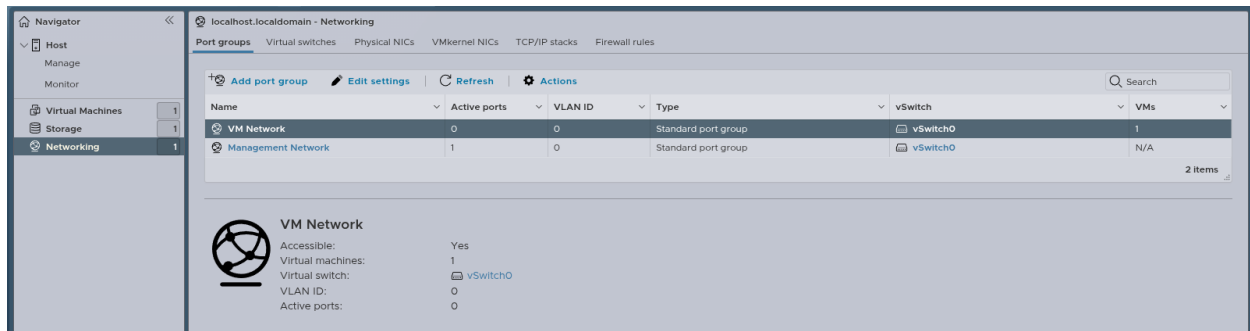
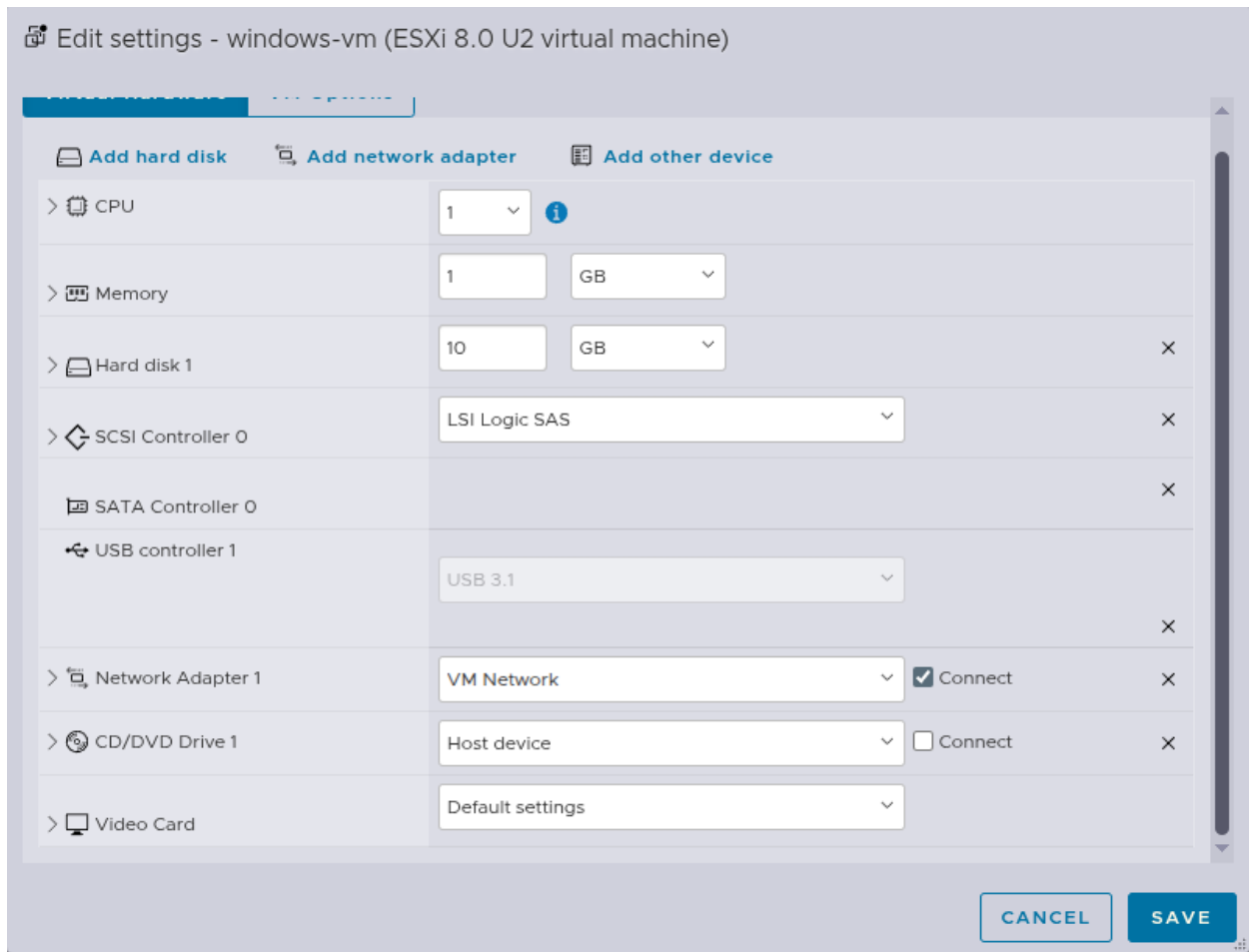
Memory: 1.95 GB

CPU: 0 MHz

MEMORY: 0 B

STORAGE: 2.64 KB

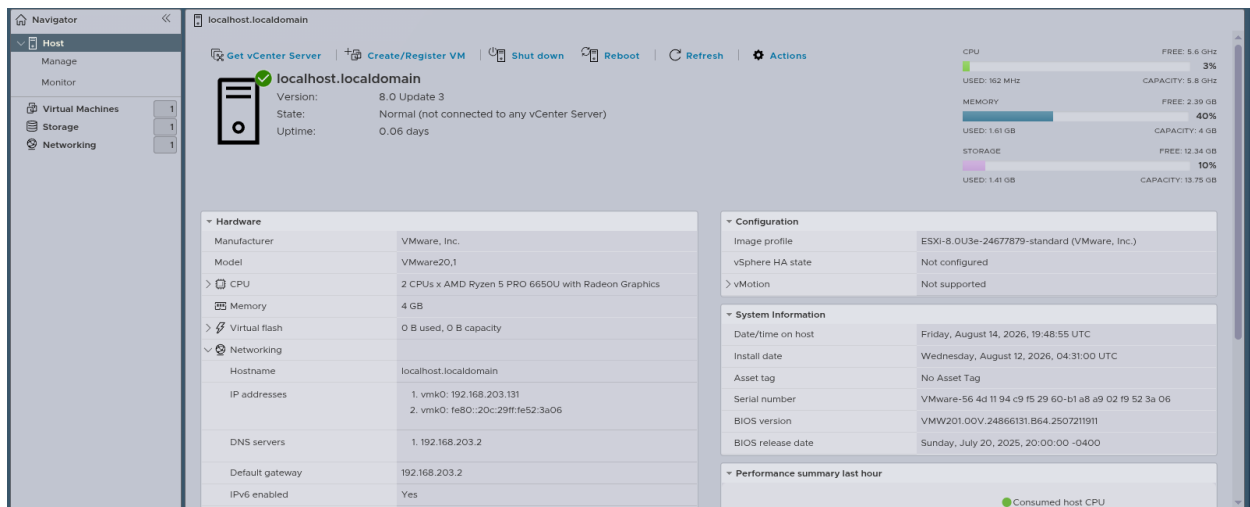
Actions -> edit settings



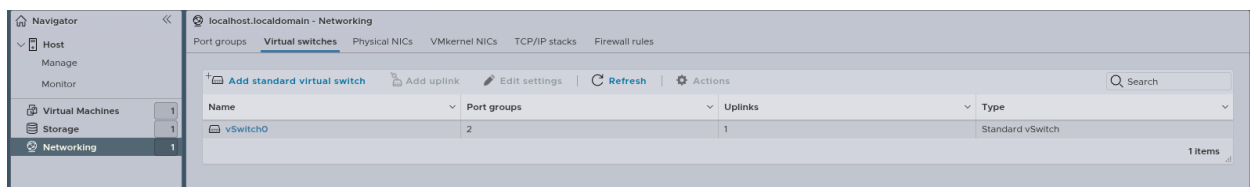
CD/DvD drive -> datastore iso file -> upload

Exploring ESXi Interface and Perform Basic Host Operations

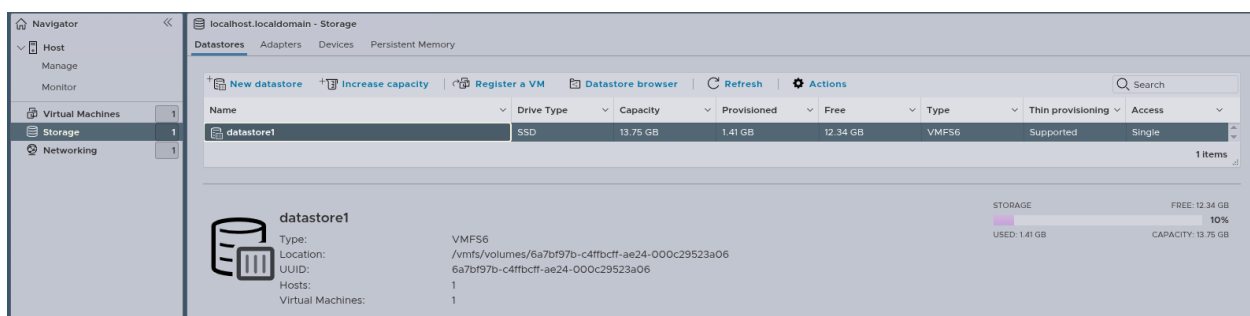
Host Summary: Displays high-level hypervisor performance metrics, hardware specs, total vs. utilized CPU and memory capacity, and overall datastore health.



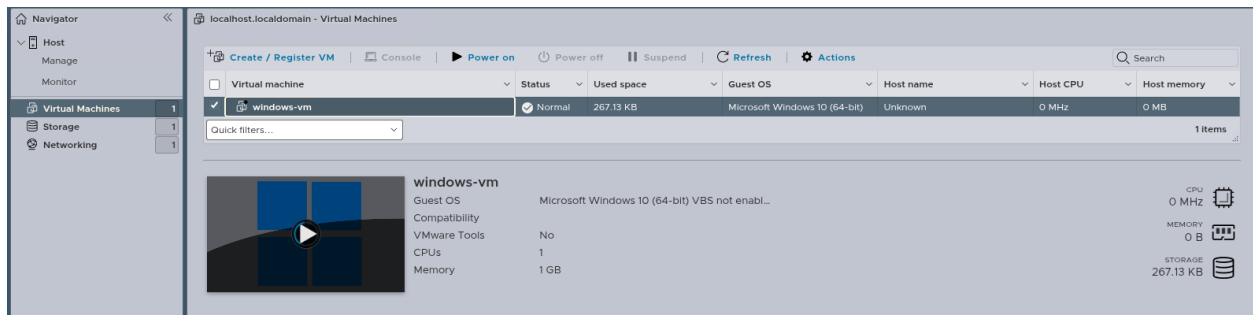
Virtual switches: Manages virtual network architecture, including standard switches (like vSwitch0) and port groups such as "VM Network" that bridge virtual machines to external networks.



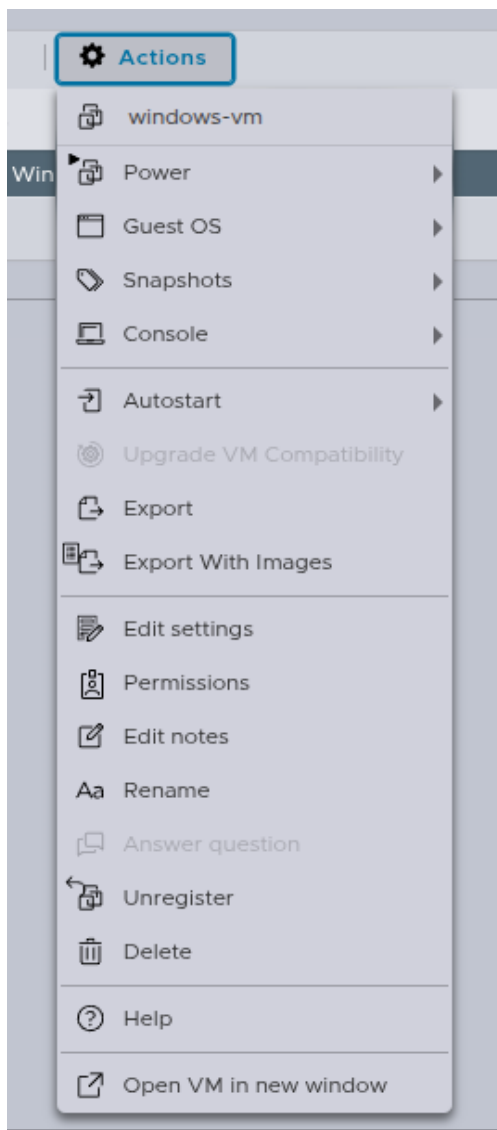
Datastore: Outlines available physical and logical storage volumes (such as datastore1), tracks free space capacity, and houses virtual machine files, configurations, and installation ISOs.



VM: all provisioned virtual machines (such as windows-vm), tracking their current power status, guest OS type, assigned memory/vCPU resources, and overall provisioning state.



Actions: Contains critical hypervisor management operations, including powering down or restarting the host, entering maintenance mode, managing system settings, and configuring host-level services.



Snapshots: Captures the current state, data, and settings of the virtual machine at a specific point in time, enabling you to revert back if an installation or update fails.

Console: Opens a remote view of the virtual machine's display so you can interact with the operating system installation or desktop interface.

Export: Downloads the virtual machine as an OVF/OVA template package for backup, archiving, or migration to another hypervisor host.

Unregister: Removes the virtual machine from the ESXi host inventory without deleting its underlying files from the datastore.

Delete: Completely removes the virtual machine and permanently erases all of its associated files and virtual disks from the datastore.

Learning outcomes

- Understand the fundamentals and importance of enterprise virtualization.
- Differentiate between physical and virtual infrastructure.
- Understand the role and architecture of VMware ESXi and its relationship with vSphere and virtual machines.
- Install and configure VMware ESXi in a virtualized lab environment.

Conclusion

Enterprise virtualization provides an efficient and flexible approach to managing IT infrastructure by allowing multiple virtual machines to operate on shared physical resources. VMware ESXi provides practical

experience with installing and configuring a hypervisor, managing host networking and storage, accessing the Host Client, and creating and configuring virtual machines. These concepts establish an important foundation for more advanced infrastructure administration and prepare for the complete infrastructure deployment in the upcoming capstone work.

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